



Intelligent Document Processing for Scalable Automation

Problem Statement:

- This Proof of Concept aimed to evaluate an AI-driven system capable of performing bilingual OCR in English and Gujarati, along with automatic document classification and data extraction. The goal was to enhance digitization efficiency, minimize manual processing effort, and enable reliable automation of document workflows in Govt. departments. The system was expected to demonstrate production-grade accuracy, usability, and speed while remaining scalable for real-world deployment.

Solution:

- A fully no-code, web-based interface was deployed to demonstrate the end-to-end document processing workflow.
- It is built on a containerized architecture, enabling flexible deployment and can facilitate large scale production
- Human-in-the-Loop capabilities were enabled, allowing users to review, validate, and correct extracted data when necessary.
- Additional validation features were included to ensure accuracy and reliability of the processed outputs.
- Integration with existing Govt. systems was tested and facilitated

Impact:

- Processed 3,000+ documents with 100% ingestion success and system uptime.
- Demonstrated extraction of 17+ fields and potential for 150× faster data extraction, saving 12–17 minutes per page.
- Delivered 96% classification accuracy with results in under 3 seconds across four categories.
- Tested four OCR pipelines for English and Gujarati (including handwritten Gujarati) and provided recommendations